Project Report

Client: American Speech-Language-Hearing Association (ASHA)

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1. Postcard Experiment

1.1 Postcard Experiment Overview

The 2025 SLP Healthcare Survey Postcard Reminders Experiment was conducted alongside the web survey administered via SurveyMonkey. Its purpose was to examine whether a postcard reminder encouraged nonresponders to return to their email and complete the survey.

Table 1. Basic Experiment Information

Date	Group	N
Feb 27	Initial web survey sent to full sample	**9,947**
Mar 13	Nonrespondents selected to receive first postcard	**5,000**
	Nonrespondents not selected to receive first postcard	4,947
Apr 10	Re-selected from Group B nonresponders, received second postcard	**2,500**

Two postcards were used in the experiment. Postcard 1 ("We want to hear from you!") highlighted the value of SLP input on topics such as salary, safety, and on-the-job challenges, informed recipients that the survey link would arrive by email on March 27, and asked them to complete the survey by April 7. Postcard 2 ("Your voice matters, and so does the work you do!") was a follow-up reminder asking recipients to check their email for the link and submit their feedback to the SLP Health Care Survey by May 5.

1.2 Data Source

The data come from the 2025 SLP Health Care Survey. ASHA used probability (nonreplacement) sampling via a stratified systematic technique to select a sample of 15,000 ASHA-certified speech-language pathologists (SLPs). The survey was fielded in the spring of 2025 and obtained an unweighted response rate of 18.6% (2,693 completed surveys from a net sample of 14,496 eligible SLPs). The postcard experiment and the analysis in this report are based only on the web survey portion of this sample, which covers 9,947 SLPs.

The data used for analysis are organized into four parts. The first three contain only anonymized ID numbers: the Full Sample (9,947 IDs), the Postcard 1 Nonresponders (5,000 IDs), and the Postcard 2 Nonresponders (2,500 IDs). The fourth, Survey Respondents with Date, contains all respondents, each with an anonymized respondent ID together with the start date (when the respondent began the survey) and the end date (when the respondent submitted the survey).

1.3 Research Goal

The overall goal is to evaluate whether sending postcard reminders to nonresponders of the web-based 2025 SLP Health Care Survey improves the response rate, and to quantify the effect of each postcard mailing.

1.4 Client Research Questions

- Q1: Estimated response rates and the differences across groups receiving 0, 1, or 2 postcards.
- Q2: The effect of the first postcard: does receiving the first postcard improve the response rate?
- Q3: The effect of the second postcard: does receiving the second postcard improve the response rate?
- Q4: Were there boosts in response to raw numbers within a week of each postcard being sent?

1.4.1 Q1: Estimated response rates and the differences across groups receiving 0, 1, or 2 postcards.

1.4.1.1 Q1 Analysis Plan

Motivation

A naive comparison of observed response rates across groups defined by the number of postcards *received* is biased by **treatment by indication**: recipients of Postcard 1 were already self-selected nonresponders at t_1 = March 13, and recipients of Postcard 2 were doubly-selected nonresponders at t_2 = April 10. The correct estimand is a *policy-level* response rate — what the overall response rate would have been had *everyone* in the sample been assigned to a given mailing regime. Randomization within each risk set makes this estimand identifiable from the observed data.

Decomposition via conditional probability

The overall probability of responding by the end of the study can be written by the law of total probability as:

$$P(R = 1 \text{ by end}) = P(R = 1 \text{ by time } a) + [1 - P(R = 1 \text{ by time } a)] \times P(R = 1 \text{ after time } a \mid \text{no response by time } a)$$

That is, the final response rate equals the pre-intervention response rate plus the probability of *not* having responded by time *a*, multiplied by the conditional post-intervention response rate *among the risk set at time a*. This identity is applied sequentially at each randomization point (t_1 = March 13; t_2 = April 10).

Five Component Rates

In terms of final postcard exposure, 4,947 subjects received 0 cards, 2,500 received exactly 1 card, and 2,500 received 2 cards. These counts describe actual final exposure status; the denominators for RR_2–RR_5 are the relevant risk sets at each intervention stage, not the final arm sizes. The analysis uses five observable component response rates defined from the full sample and successive post-randomization risk sets (Table 1):

Table 2. Component Response Rates (RR_1–RR_5)

Rate	Description	Estimate	Analytic set
RR_1	Response before Mar 13 (pre-randomization)	6.23%	Full sample (N = 9,947)
RR_2	Eventual response after t_1, 0-card arm	7.71%	Nonresponders at t_1 randomized to no first postcard
RR_3	Response between t_1 and t_2, 1st-card arm	3.41%	Nonresponders at t_1 randomized to first postcard
RR_4	Response after t_2, 1-card-only arm	5.84%	Nonresponders at t_2 in the one-card-only arm
RR_5	Response after t_2, 2nd-card arm	6.04%	Nonresponders at t_2 in the second-postcard arm

Policy estimators

The three counterfactual response rates are constructed as follows:

$$P(RR \mid card = 0) = RR_1 + (1 - RR_1) \times RR_2$$
$$P(RR \mid card = 1) = RR_1 + (1 - RR_1) \times [RR_3 + (1 - RR_3) \times RR_4]$$
$$P(RR \mid card = 2) = RR_1 + (1 - RR_1) \times [RR_3 + (1 - RR_3) \times RR_5]$$

The no-card estimator uses RR_2 as the post-t_1 rate. The one-card estimator replaces it with the sequential combination of the first-mailing effect (RR_3) and the residual response rate among those still non-responding at t_2 (RR_4). The two-card estimator substitutes RR_5 for RR_4 to capture the additional gain from Postcard 2. Because the covariance structure across these estimators is complex, uncertainty is quantified via **stratified bootstrap** (B = 2,000 resamples), resampling within each final card group (0, 1, 2) to preserve the allocation proportions. Bootstrap SEs and 2.5th/97.5th percentile confidence intervals are reported.

1.4.1.2 Q1-Q3 Results

Pre-randomization baseline

RR_1 = 6.23%. About 6% of the full web sample responded before the first randomization date (March 13). This rate is pooled across all three eventual groups, as intended, since mailing assignment had not yet occurred.

Effect of Postcard 1

Table 2 presents the three policy-estimated response rates with bootstrap SEs and 95% CIs. Under the no-postcard policy, the estimated overall response rate is **13.5%** (95% CI: 12.6%–14.3%). Under the one-postcard policy it rises to **14.7%** (95% CI: 13.6%–15.8%), corresponding to an estimated increase of **+1.26 percentage points** (95% CI: 0.00–2.51 pp; see Table 3 and Figure 1). The 95% CI excludes zero (pre-rounding lower bound: +0.004 pp), indicating a statistically significant, if modest, positive effect of the first postcard on eventual response.

Table 3. Policy-estimated response rates (bootstrap, B = 2,000)

Setting	Estimate	Bootstrap SE	CI Lower 95%	CI Upper 95%
No postcard	13.50%	0.44 pp	12.60%	14.30%
Up to 1 postcard	14.70%	0.54 pp	13.60%	15.80%
Up to 2 postcards	14.90%	0.54 pp	13.90%	16.00%

Effect of Postcard 2

The two-postcard policy yields an estimated rate of **14.9%** (95% CI: 13.9%–16.0%), only **+0.19 pp** above the one-postcard rate (95% CI: −1.06 to +1.43 pp). This interval includes zero and provides no evidence of a meaningful additional effect from the second mailing. The near-identical values of RR_4 (5.84%) and RR_5 (6.04%) suggest little to no additional response gain from Postcard 2 among those who had already received and not responded to the first mailing.

Table 4. Pairwise differences in policy-estimated response rates (bootstrap, B = 2,000)

Comparison	Difference	SE	CI Lower	CI Upper	Significant
1 postcard vs. no postcard	+1.26 pp	0.63 pp	0.00 pp	+2.51 pp	Yes
2 postcards vs. 1 postcard	+0.19 pp	0.62 pp	-1.06 pp	+1.43 pp	No
2 postcards vs. no postcard	+1.44 pp	0.62 pp	+0.21 pp	+2.69 pp	Yes

Note: "Significant" = 95% CI excludes zero (before rounding). pp = percentage points.

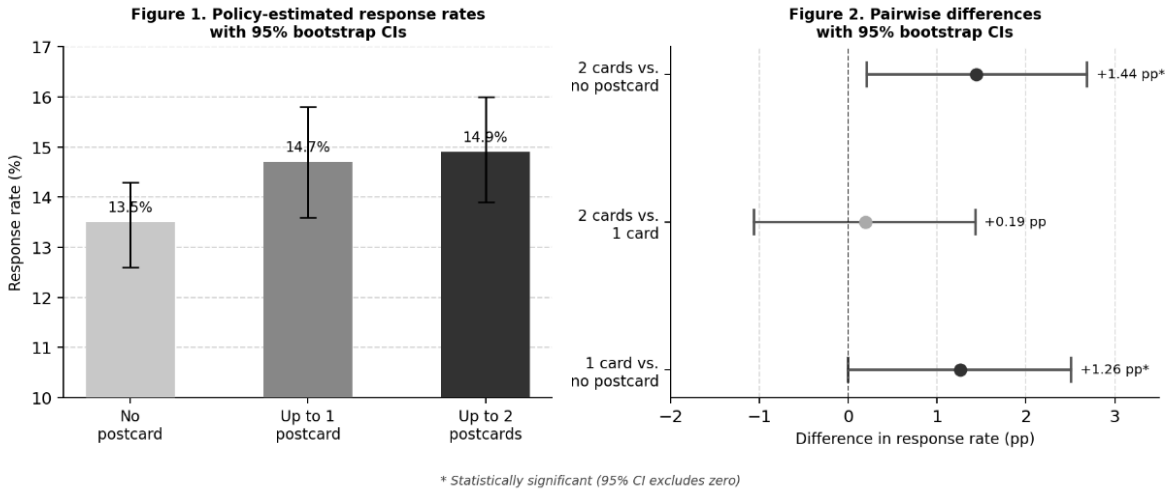

Figure 1. *Left: Policy-estimated response rates with 95% bootstrap CIs. Right: Pairwise differences with 95% bootstrap CIs (* = CI excludes zero before rounding).*

Cumulative effect of any mailing

The total gain from a two-postcard protocol versus no protocol is **+1.44 pp** (95% CI: 0.21–2.69 pp), which is statistically significant and driven almost entirely by Postcard 1.

Recommendations for Future Survey Design

The evidence supports sending Postcard 1 but not Postcard 2 in its current form. The first mailing provides a statistically significant, if modest, boost to response rates at the cost of contacting approximately half the sample; the second mailing to remaining nonresponders yields no detectable benefit. If resources for a second contact effort are available, they would be better redirected toward redesigning the reminder — altering timing, message framing, or delivery channel (e.g., a follow-up email rather than a physical postcard).

1.4.2 Q4: Were there boosts in response raw numbers within a week of each postcard being sent?

To examine whether postcard reminders produced short-term increases in survey responses, we analyzed daily response counts around each postcard mailing date.

Step 1. Generate daily response counts

For each calendar date (*d*), we calculated the number of completed surveys submitted on that date within each mailing group. This produced a daily time series of response counts for Group A, Group B, Group C, and Group B_only. Missing dates were filled with zeros so that each group had a complete daily sequence across the whole period.

Step 2. Define time windows around each postcard mailing

Two postcard mailings were evaluated. The first postcard was sent on March 13, 2025, and the second postcard was sent on April 10, 2025. For each mailing, we defined a 7-day post-mailing window to capture short-term response activity immediately after the postcard was sent. We also defined a corresponding 7-day pre-mailing window to provide a baseline for comparison.

For the first postcard:

- Pre1 window: March 7 to March 13
- Post1 window: March 14 to March 20

For the second postcard:

- Pre2 window: April 4 to April 10
- Post2 window: April 11 to April 17

These windows were chosen to isolate short-term response patterns around the postcard dates while avoiding overlap with the later email reminders shown in the timeline.

Step 3. Calculate average daily responses within each window

Within each pre- and post-mailing window, we calculated the average daily number of responses for the relevant treatment and comparison groups.

For the first postcard, the comparison was between:

- Group A: respondents who did not receive a postcard
- Group B: respondents who received the first postcard

For the second postcard, the comparison was between:

- Group B_only: respondents who received only the first postcard

- Group C: respondents who received both postcards

The mean daily response count was calculated as the average of the seven daily counts within each window.

Step 4. Compute the pre-post change ("boost")

For each group, we defined the short-term boost as the change in average daily responses from the pre-window to the post-window:

$$Boost = Mean\ After - Mean\ Before$$

For the **first postcard**, define the pre-post change in each group as:

$$Boost_{1,A} = MeanAfter_{1,A} - MeanBefore_{1,A}$$
$$Boost_{1,B} = MeanAfter_{1,B} - MeanBefore_{1,B}$$

For the **second postcard**, define:

$$Boost_{2,B_{only}} = MeanAfter_{2,B_{only}} - MeanBefore_{2,B_{only}}$$
$$Boost_{2,C} = MeanAfter_{2,C} - MeanBefore_{2,C}$$

A positive boost indicates that average daily responses increased after the postcard mailing, while a negative boost indicates that responses declined.

Step 5. Calculate the Difference-in-Differences (DiD) estimate

Because response activity generally declined over time during the survey period, a simple pre-post comparison could confound postcard effects with the broader downward trend. To address this, we calculated a difference-in-differences (DiD) estimate for each postcard mailing. For the first postcard, the DiD estimate was:

$$DiD_1 = Boost_{1,B} - Boost_{1,A}$$

where Group B served as the treatment group and Group A served as the comparison group. For the second postcard, the DiD estimate was:

$$DiD_2 = Boost_{2,C} - Boost_{2,B_{only}}$$

where Group C served as the treatment group and Group B_only served as the comparison group. A positive DiD value indicates that the postcard group experienced a more favorable change in daily responses than the comparison group over the same period.

Step 6. Plot daily response trends

To complement the window-based calculations, we plotted daily response counts over time for four groups. These plots provide a visual check on whether short-term response spikes occurred after postcard mailings.

Results

Table 5. Average Daily Responses Before and After Each Postcard Mailing

Postcard	Group	Mean Before	Mean After	Boost (After - Before)
Postcard 1 (Mar 13)	Group A (no postcard)	1.86	0.14	-1.71
Postcard 1 (Mar 13)	Group B (received first postcard)	1.43	1.00	-0.43

Postcard 2 (Apr 10)	Group B_only (received first postcard only)	0.29	0.14	-0.14
Postcard 2 (Apr 10)	Group C (received two postcards)	0.00	0.57	+0.57

Table 6. Difference-in-Differences Estimates for Postcard Effects

Postcard	Treatment Group	Comparison Group	Treatment Boost	Comparison Boost	DiD Effect
Postcard 1	Group B	Group A	-0.43	-1.71	+1.29
Postcard 2	Group C	Group B_only	+0.57	-0.14	+0.71

For the first postcard mailing, average daily responses decreased in both groups during the week after mailing, but the decline was smaller among respondents who received the postcard. Group A declined from 1.86 to 0.14 responses per day, while Group B declined from 1.43 to 1.00. This produced a difference-in-differences estimate of +1.29 daily responses, indicating a positive relative effect of the first postcard.

For the second postcard mailing, Group B_only declined slightly from 0.29 to 0.14 responses per day, whereas Group C increased from 0.00 to 0.57. The resulting difference-in-differences estimate was +0.71 daily responses, suggesting an additional short-term gain associated with the second postcard reminder.

These results show that, although response counts generally declined over time, the decline was smaller among first-postcard recipients than among nonrecipients, and the second-postcard group showed a modest increase relative to the one-postcard-only group.

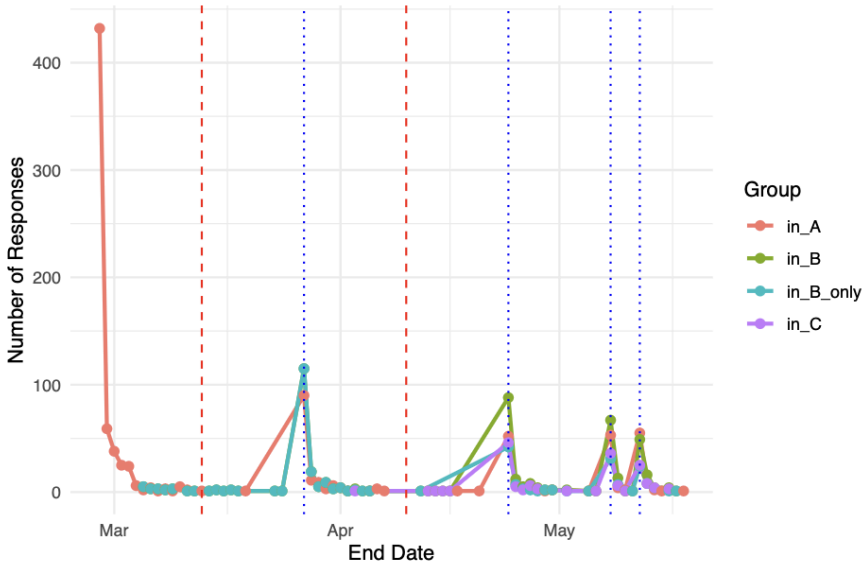

Figure 2. *Daily Response Counts by Four Groups*

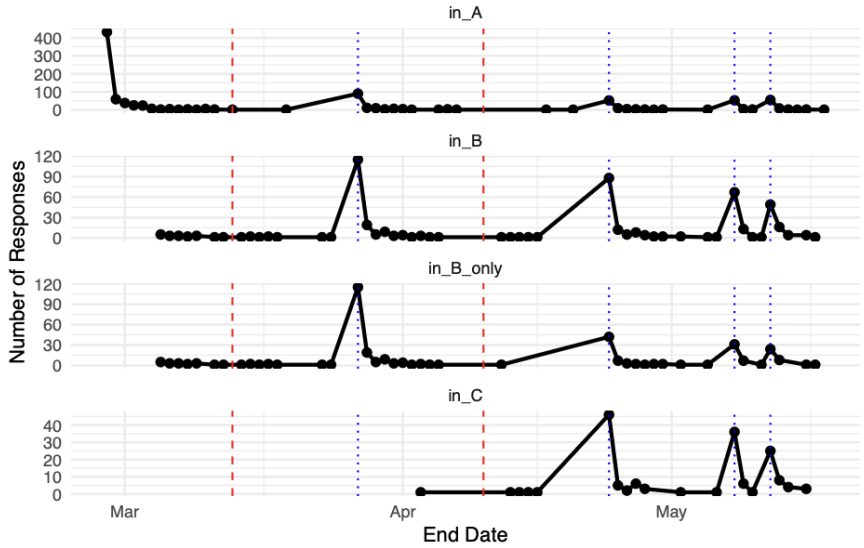

Figure 3. *Daily Response Counts for Four Groups (Separate Panels)*

In daily response counts plots, postcard mailing dates are marked with red dashed lines, and email reminder dates are marked with blue dotted lines. The time-series patterns suggest that postcard mailings were followed by small short-term increases in responses, although the absolute changes were close to zero. In contrast, larger visible spikes occurred after the email reminders, suggesting that email reminders may have generated stronger immediate increases in participation than postcard mailings.

Recommendations for Future Survey Design

From a practical standpoint, if postcard delivery involves substantial printing and mailing costs, email reminders may be a more efficient alternative for boosting short-term participation. If postcards continue to be used, their effectiveness might be improved by adding a direct survey access tool, such as a QR code, so that recipients can immediately scan and complete the survey. This could reduce response burden and make the postcard function more like an email reminder with a direct survey link.

2. Survey Mode Experiment

2.1 Brief Introduction

Survey mode refers to the medium through which a questionnaire is delivered to respondents. As one of the most consequential design decisions in survey methodology, mode selection directly affects respondent engagement, accessibility, and overall response rates.
ASHA has conducted multiple mode experiments across different survey years, consistently comparing postal mail against web-based administration (SurveyMonkey). Across these experiments, postal mail has yielded higher response rates than web-based modes, motivating a more rigorous investigation into the nature and consistency of this mode effect. This section focuses on two recent surveys (the 2024 SLP Schools Survey and the 2025 SLP Health Care Survey) to quantify the overall mode effect, identify whether certain subgroups are more sensitive to survey mode, and assess whether differential response by mode introduces meaningful composition bias.

2.2 Data Source

- **2024 SLP Schools Survey.** The 2024 SLP Schools Survey was administered to a sample of 15,000 CCC-SLPs working in school settings, stratified by state. The survey employed a mixed-mode design in which 50% of respondents were randomly assigned to receive a postal mail questionnaire and 50% received a web-based survey via SurveyMonkey. Among SLP respondents, postal mail yielded a response rate of 29.9% and SurveyMonkey yielded 21.4%. Key subgroup variables available in this dataset include survey mode, respondent age, facility type, employment status, and state (recoded into four U.S. Census regions: Northeast, Midwest, South, and West).
- **2025 SLP Health Care Survey.** The 2025 SLP Health Care Survey targeted SLPs working in health care settings. Similar to the 2024 survey, it employed a postal mail versus SurveyMonkey mixed-mode design. The overall response rates were 28.6% for postal mail and 13.5% for SurveyMonkey. Subgroup variables available for this survey include job function, facility type, and state.

2.3 Methods

The analysis addresses three research questions: (1) How does survey mode affect overall response rates? (2) Do subgroups differ in their sensitivity to survey mode? (3) Do different modes change response propensity across subgroups?

Three complementary analytic approaches were applied consistently across both surveys and all subgroups.

1) **Response rate comparison.** For each subgroup, response rates were calculated separately by mode and the difference (Δ = RR Mail − RR Web) was computed to quantify the mode advantage.

2) **Chi-square tests.** Pearson's chi-square test was applied to 2×2 contingency tables (mode × responded/not responded) within each subgroup to assess whether the observed mode differences were statistically significant.
3) **Logistic regression with Likelihood Ratio Test (LRT).** To formally test whether the mode effect varies across subgroups, an interaction model was estimated:

$$log(\frac{p}{1-p}) = \beta_0 + \beta_1 Mode + \beta_2 Subgroup + \beta_3 (Mode \times Subgroup)$$

The coefficient β_3 captures the differential mode effect across subgroup levels. A Likelihood Ratio Test (LRT) comparing the interaction model to a main-effects-only model was used to determine whether adding the interaction term significantly improved model fit. A significant LRT ($p < .05$) indicates that the mode effect differs meaningfully by subgroup, suggesting that subgroup-specific mode strategies may be warranted.

2.4 Results: 2024 SLP Schools Survey

2.4.1 Overall Mode Effect

Across all subgroups, postal mail consistently achieves higher response rates than SurveyMonkey. The magnitude of this difference varies substantially across demographic and contextual factors.

Subgroup 1: Employment Status

Response rates differ by employment status, but the mode effect remains consistent.

- Full-time: Mail (29.8%) vs Web (21.8%)
- Part-time: Mail (31.0%) vs Web (16.8%)

The mode gap is larger among part-time employees, indicating that web surveys may be less effective for this group.

Subgroup 2: Facility Type

Response rates vary significantly across facility types.

- Highest response rates are observed in elementary and administrative settings.
- Lower response rates appear in pre-elementary and "other" categories.

Across all facility types, postal mail outperforms web mode, with differences often exceeding **8–10 percentage points**.

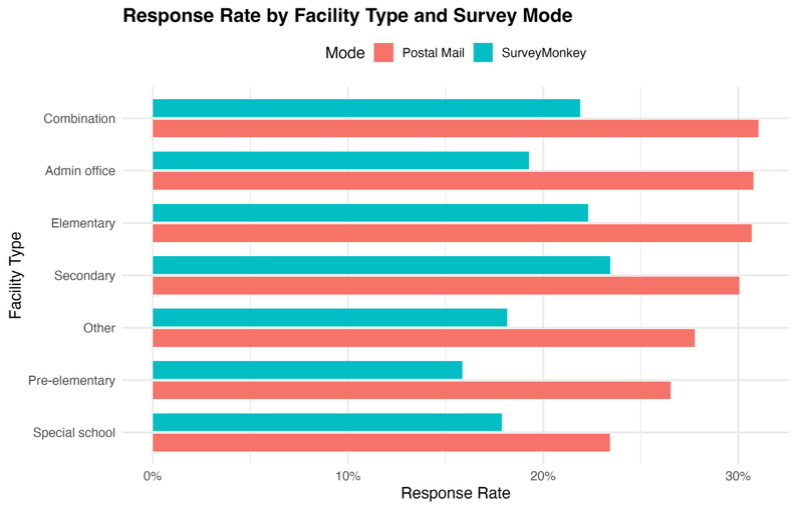

Figure 4. *Response Rate by Facility and Survey Mode*

Subgroup 3: States and Region

Regional variation is present but does not eliminate the mode effect.

- Midwest and West show relatively higher response rates.
- Northeast and South show comparatively lower rates.

However, in all regions, mail response rates exceed web response rates by approximately **5–13 percentage points**.

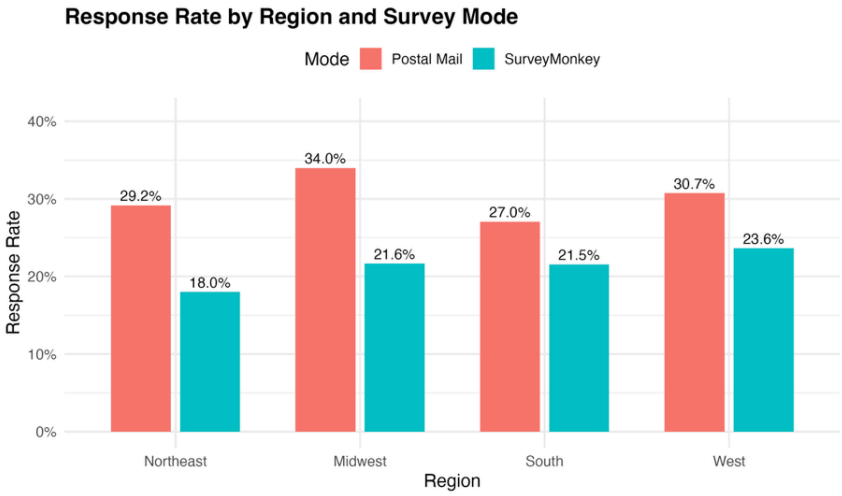

Figure 5. *Response Rate by Region and Survey Mode*

States were recoded into four U.S. Census regions to enable region-level analysis. Response rates by region and mode are presented in Table X and Figure X.

Table 7: Response Rates by Census Region and Mode (2024 SLP Schools Survey)

Region	RR Mail	RR Web	Δ (Mail − Web)
Midwest	34.0%	21.6%	+12.4 pp
Northeast	29.2%	18.0%	+11.2 pp
South	27.0%	21.5%	+5.5 pp
West	30.7%	23.6%	+7.1 pp

However, the LRT for the Mode × Region interaction term yielded ΔDeviance = 1.16, df = 3, p = .762, indicating that adding the interaction did not significantly improve model fit. This means the magnitude of the mail advantage was broadly uniform across Census regions — no region-specific mode strategies are statistically justified based on these data.

To assess whether differential response by region introduced composition bias, the regional composition of the respondent sample was compared against the eligible sample (Table X and Figure X). The differences were minimal in absolute terms: the Midwest was slightly over-represented (+2.1 pp) and the Northeast slightly under-represented (−1.8 pp), while the South (−1.6 pp) and West (+1.4 pp) showed similarly small deviations. These small differences indicate that mode-driven differential nonresponse did not meaningfully distort the regional composition of the 2024 respondent sample.

Table 8: Regional Composition: Eligible Sample vs. Respondent Sample (2024 SLP Schools Survey)

Census Region	Eligible N	Eligible %	Respondent N	Respondent %	Difference (pp)
Midwest	3,503	23.9%	973	25.9%	+2.1
Northeast	3,375	23.0%	794	21.2%	−1.8
South	4,657	31.7%	1,131	30.2%	−1.6
West	3,138	21.4%	853	22.7%	+1.4

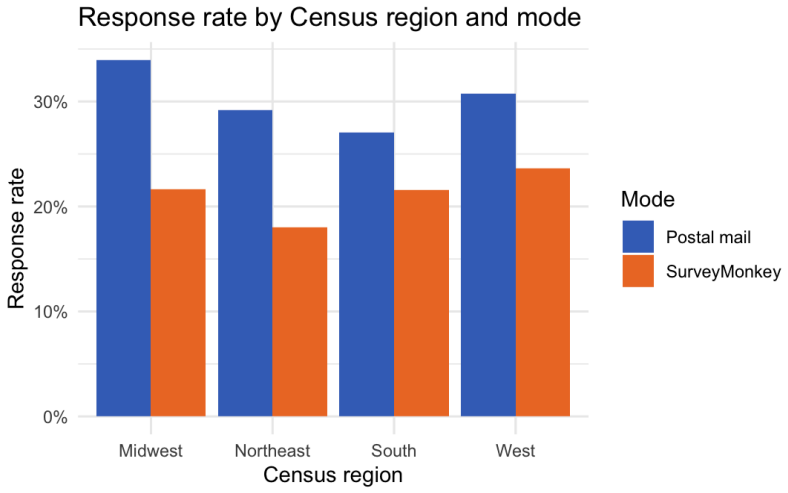

Figure 6. Response Rate by Census Region and Mode

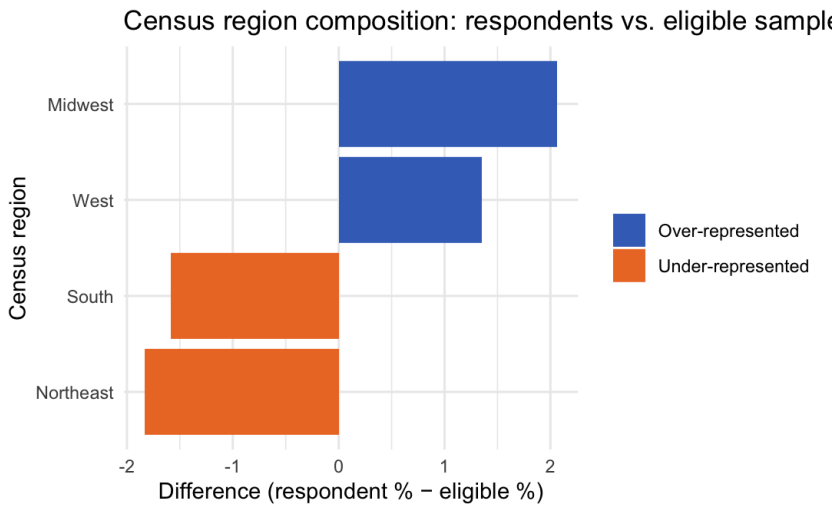

Figure 7. Census Region Composition: Respondents vs. Eligible Sample

From a practical perspective, these findings suggest that region-specific mode strategies are unnecessary for the 2024 SLP Schools Survey. The same general mail-first pattern applies uniformly across the country, and the respondent sample composition closely mirrors the eligible sample composition regardless of mode.

Subgroup 4: Age Group

Age shows the strongest interaction with survey mode.

- Mail response increases steadily with age, peaking in older groups.
- Web response remains relatively flat and drops sharply for the 65+ group.

The largest mode difference is observed for respondents aged 65+, where mail outperforms web by over **20 percentage points**.

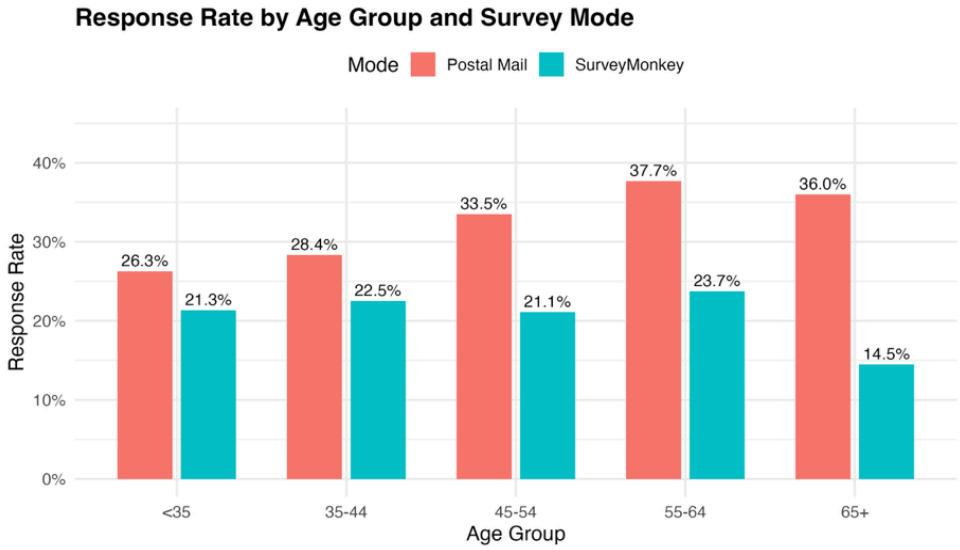

Figure 8. Response Rate by Age Group and Survey Mode

2.4.2 Mode Advantage Summary

To summarize the mode effect across subgroups, the difference between mail and web response rates was computed.

The largest advantages for mail are observed in:

- Age 65+
- Part-time employees
- Midwest and Northeast regions

Smaller differences are observed among younger respondents and in the South.

Association Between Mode and Response

A chi-square test of independence was conducted to examine the relationship between survey mode and response status. The test is highly significant ($\chi^2 = 140.2$, df = 1, $p < 2.2e{-}16$), indicating a strong association between survey mode and response. Respondents are significantly more likely to respond under postal mail compared to SurveyMonkey. The magnitude of the test statistic suggests that this is not a marginal effect but a substantial difference in response behavior across modes.

$$\chi^2(1) = 140.2,\ p < 0.001$$

To further assess whether this relationship holds within subgroups, additional chi-square tests were conducted separately by employment status, age group, geographic region, and facility type. These tests are statistically significant across nearly all subgroup levels, confirming that the association between mode and response is robust across the population. The only exceptions are administrative offices and special school settings, where differences are not statistically significant, likely due to smaller sample sizes or weaker mode differentiation in these contexts.

Chi Square Table

To evaluate whether the effect of survey mode on response is statistically significant within subgroups, Pearson's chi-square tests were conducted separately for each level of employment status, age group, geographic region, and facility type. The results are presented in Table X.

Table 9. Chi-square Tests of Mode × Response by Subgroup (2024 SLP Schools Survey)

Subgroup	Level	Chi-square	df	p-value	Significant
Employment	Employed full time	104.0	1	1.84e-24	Yes
Employment	Employed part time	41.4	1	1.21e-10	Yes
Age	<35	9.74	1	1.80e-03	Yes
Age	35–44	16.7	1	4.31e-05	Yes
Age	45–54	72.3	1	1.81e-17	Yes
Age	55–64	36.2	1	1.77e-09	Yes

Age	65+	24.4	1	7.66e-07	Yes
Region	Midwest	62.3	1	3.00e-15	Yes
Region	Northeast	53.6	1	2.41e-13	Yes
Region	South	17.2	1	3.28e-05	Yes
Region	West	20.0	1	7.65e-06	Yes
Facility	Admin office	2.85	1	9.12e-02	No
Facility	Combination	43.6	1	3.96e-11	Yes
Facility	Elementary	55.5	1	9.44e-14	Yes
Facility	Other	5.62	1	1.77e-02	Yes
Facility	Pre-elementary	25.3	1	4.94e-07	Yes
Facility	Secondary	5.19	1	2.27e-02	Yes
Facility	Special school	1.76	1	1.85e-01	No

Across employment, age, and regional subgroups, the chi-square tests are uniformly statistically significant (all $p < 0.001$), indicating a strong and consistent association between survey mode and response. In both full-time and part-time groups, respondents are significantly more likely to respond under postal mail than SurveyMonkey. Similarly, all age groups from younger respondents (<35) to older respondents (65+) show statistically significant mode effects, confirming that the mail advantage persists across the age distribution. Regional results are also consistent, with all four Census regions exhibiting significant differences between mail and web modes.

Facility-level results are largely consistent with this pattern but show some heterogeneity. Most facility types including elementary, combination, and pre-elementary settings show statistically significant mode effects. However, two facility categories administrative offices ($p = 0.091$) and special schools ($p = 0.185$) do not show statistically significant differences. These non-significant results suggest that in these specific settings, the difference between mail and web response rates may be smaller or more variable, potentially reflecting smaller sample sizes or more homogeneous response behavior across modes.

Overall, these findings confirm that the effect of survey mode is statistically robust across nearly all subgroups. The consistent significance across employment, age, and region reinforces the conclusion that postal mail systematically outperforms web-based administration. The few non-significant facility categories do not materially alter this conclusion but suggest that the strength of the mode effect may vary slightly depending on context.

2.4.3 Logistic Regression Results

To assess the effect of survey mode while controlling for demographic and contextual variables, a logistic regression model was estimated.

Mode Effect

The coefficient for SurveyMonkey is negative and highly significant:

· Estimate: -0.468

· $p < 0.001$

· Odds ratio: 0.63

This implies that, holding all other variables constant, the odds of responding via SurveyMonkey are approximately **37% lower** than via postal mail. This confirms a strong and persistent mode effect.

Age Effects

Age is a significant predictor of response.

- Ages 35–44: OR = 1.13 ($p = 0.032$)
- Ages 45–54: OR = 1.24 ($p < 0.001$)
- Ages 55–64: OR = 1.43 ($p < 0.001$)
- Ages 65+: Not statistically significant

Response likelihood increases with age up to 55–64. Although the 65+ category is not statistically significant in the regression, subgroup chi-square tests show a strong mode effect within this group, suggesting that response behavior in this category may be more sensitive to survey mode than captured by the main-effects model.

Regional Effects

Regional differences are statistically significant for some regions:

- Northeast: OR = 0.81 ($p < 0.001$)
- South: OR = 0.82 ($p < 0.001$)
- West: Not significant

Respondents in the Northeast and South have lower odds of responding compared to the reference region, while the West does not differ significantly. However, subgroup chi-square tests show that the mode effect remains statistically significant across all regions, indicating that the mail advantage is consistent nationwide.

Employment Effects

Employment status also influences response:

· Part-time employment: OR = 0.81 ($p < 0.001$)

Part-time respondents are significantly less likely to respond compared to full-time respondents. Subgroup chi-square tests further confirm that the mode effect is statistically significant within both employment groups, with a particularly strong effect among part-time respondents.

Facility Type Effects

Facility type does not show statistically significant differences after controlling for other variables. While some categories (e.g., elementary, combination, secondary) have positive coefficients, none are statistically significant at conventional levels.

However, subgroup chi-square results indicate that mode effects are statistically significant in most facility types, with the exception of administrative offices and special schools. This suggests that while facility type does not independently predict response in the regression model, it may still influence how survey mode affects response behavior in specific contexts.

Model Fit

- Null deviance: 15159
- Residual deviance: 14921
- AIC: 14953

The reduction in deviance indicates that the model improves fit relative to the null model, though a substantial portion of variation remains unexplained, which is expected in response propensity models.

Interpretation

Both the chi-square test and logistic regression provide consistent evidence that:

1. Survey mode is a strong and statistically significant determinant of response
2. SurveyMonkey substantially reduces response likelihood relative to postal mail
3. Age and employment status are key drivers of response heterogeneity
4. Regional effects exist but are secondary to mode effects
5. Facility type has limited independent influence after adjustment

These results confirm that the differences observed in the descriptive analysis reflect a true mode effect rather than compositional differences in the sample.

Link to Visual Evidence

The inferential findings align closely with the descriptive patterns:

- Larger mail advantage for older age groups
- Strong gaps for part-time employees
- Consistent mail advantage across regions

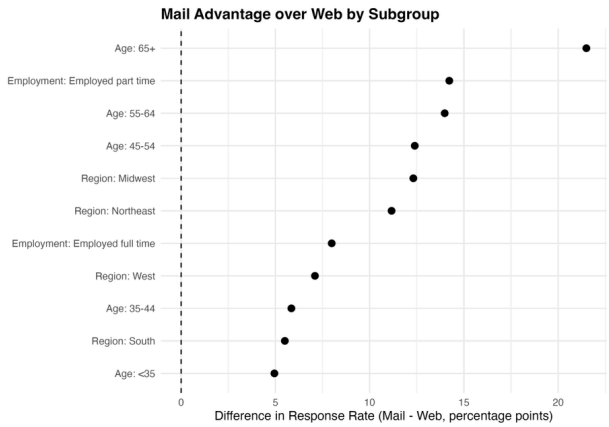

Figure 9. Mail Advantage over Web by Subgroup

Recommendations

Survey Mode Strategy

Postal mail should remain the primary mode of data collection. Both descriptive and inferential results show that SurveyMonkey significantly reduces response likelihood (odds ratio ≈ 0.63), with consistent disadvantages across all subgroups.

A mixed-mode design can still be used, but web should function as a **supplement rather than a replacement** for mail. In particular, reliance on web-only approaches would likely lead to substantial nonresponse, especially among older respondents and part-time employees.

Targeted Design for Subgroups

Survey design should account for subgroup differences in response behavior.

- Older respondents (especially 55+) respond much more reliably via mail and should be prioritized in mail-based outreach.
- Part-time employees exhibit lower response rates overall, particularly in web mode, suggesting that additional engagement strategies may be needed for this group.
- Regional differences exist but do not justify region-specific designs, as the mail advantage is consistent nationwide.

Overall, subgroup variation should inform emphasis in outreach rather than fundamentally different survey designs.

2.4.3 Results: 2025 Healthcare Survey

Subgroup 1: Employment Status

Among full-time employees, the mail response rate was 26.5% compared to 13.4% for web, a difference of 13.1 percentage points. Among part-time employees, the mail response rate was 32.3% compared to 11.7% for web, a difference of 20.6 percentage points. Mail consistently outperformed web in both employment groups, and the gap was notably larger for part-time employees.

Table 10. Response rates by employment status and survey mode.

Employment Status	n (Mail)	RR Mail	n (Web)	RR Web
Employed Full Time	3,804	26.5%	7,611	13.4%
Employed Part Time	1,183	32.3%	2,374	11.7%

Chi-square tests confirmed that the mode effect was statistically significant within each employment group. For full-time employees, $\chi^2(1) = 297.9$, $p < .001$; for part-time employees, $\chi^2(1) = 221.6$, $p < .001$. These results indicate that within each employment subgroup, response rates differed significantly between mail and web modes.

A logistic regression model with a Mode × Employment Status interaction was fitted and compared against a main-effects-only model via LRT. The LRT was statistically significant ($\chi^2(1) = 18.55$, $p < .001$), indicating that the mode effect on response propensity differed significantly across employment groups. The interaction odds ratio was 1.55 (95% CI: [1.27, 1.89]), meaning the mail advantage in odds of responding was approximately 55% larger for part-time employees than for full-time employees. Predicted probabilities derived from the interaction model matched the observed response rates closely: full-time employees showed a Mail−Web difference of 13.1 percentage points, while part-time employees showed a difference of 20.6 percentage points.

Mail outperforms web in both employment groups, with a larger advantage for part-time employees. A web-first design could underrepresent part-time workers, so mail should be retained, especially when their representation matters.

Subgroup 2: Job Function

Response rates varied across the four job function subgroups. Practice and Supervision showed rates of 28.2% (mail) and 12.9% (web), a difference of 15.3 percentage points. Administration and Leadership showed 28.2% (mail) and 16.4% (web), a difference of 11.8 percentage points. Other showed 23.1% (mail) and 11.5% (web), a difference of 11.6 percentage points. Academic and Research was the only group where mail did not outperform web: 13.0% (mail) vs. 18.4% (web), a difference of −5.4 percentage points, though this group had a small sample size (n = 61 total).

Table 11. Response rates by job function and survey mode.

Job Function	n (Mail)	RR Mail	n (Web)	RR Web
Practice and Supervision	4,456	28.2%	8,994	12.9%
Administration and Leadership	248	28.2%	420	16.4%
Other	273	23.1%	548	11.5%
Academic and Research	23	13.0%	38	18.4%

Chi-square tests within each job function subgroup yielded mixed results. The mode effect was statistically significant for Practice and Supervision ($\chi^2(1) = 474.5$, $p < .001$), Administration and Leadership ($\chi^2(1) = 13.2$, $p < .001$), and Other ($\chi^2(1) = 18.8$, $p < .001$). For Academic and Research, the mode effect was not statistically significant ($\chi^2(1) = 0.30$, $p = .582$), likely due in part to the very small sample size in this subgroup.

A logistic regression model including a Mode × Job Function interaction term was compared to a main-effects model via LRT. The interaction was not statistically significant ($\chi^2(3) = 6.04$, $p = .110$). The overall mail advantage remained strong and consistent across job functions (overall OR = 2.66, 95% CI: [2.43, 2.91], $p < .001$). None of the individual Mode × Job Function interaction terms reached conventional significance: Administration and Leadership (OR = 0.75, $p = .153$), Other (OR = 0.87, $p = .488$), and Academic and Research (OR = 0.25, $p = .064$, marginally non-significant but based on very few observations). The predicted Mail−Web probability differences from the interaction model were 15.3 pp for Practice and Supervision, 11.8 pp for Administration and Leadership, 11.6 pp for Other, and −5.4 pp for Academic and Research.

Mail outperforms web across all major job function subgroups, and the LRT confirms that the mode advantage does not differ significantly across groups. No subgroup-specific mode strategy is warranted based on job function alone. The descriptive reversal observed in Academic and Research (where web yielded a nominally higher rate) should be interpreted cautiously given the very small cell sizes (n = 61 total) and the non-significant chi-square result for that group; it does not provide sufficient evidence to recommend a web-preferred approach for academic respondents. A uniform mail-inclusive strategy is appropriate across all job function categories.

Subgroup 3: Facility

There are 6 facility categories: general hospital, home health, outpatient clinic, pediatric hospital, rehabilitation hospital, and skilled nursing.

Response rates were higher under Mail than Web across all facility types. The size of the mail advantage varied by facility, with the largest gap observed in home health (+19.2 percentage points) and the smallest in skilled nursing (+11.7 percentage points). The remaining facility types fell in between, including rehabilitation hospitals (+16.7 points), general hospitals (+16.3 points), pediatric hospitals (+16.0 points), and outpatient clinics (+12.6 points).

Table 12. Response Rates by Facility Type and Mode

Facility Type	RR Mail	RR Web	Δ (Mail − Web)
General hospital	32.4%	16.1%	+16.3%
Home health	29.7%	10.5%	+19.2%
Outpatient clinic	23.0%	10.4%	+12.6%
Pediatric hospital	29.0%	13.0%	+16.0%
Rehab hospital	31.4%	14.7%	+16.7%
Skilled nursing	26.4%	14.7%	+11.7%

Chi-square tests further confirm that the mail advantage was present across all workplace settings (all p-values < .001).

However, the logistic regression interaction analysis showed that the **size** of the mode effect was not completely uniform across facility types. The likelihood ratio test comparing the main-effects model to the interaction model was statistically significant, with ΔDeviance = 12.94, df = 5, p = .024, indicating evidence that the mode effect varied by facility.

Table 13. Logistic Regression Interaction Model for Facility Type

Reference categories: Web mode and General Hospital

Term	OR	95% CI	z	p
Mode: Mail	2.50	[2.07, 3.01]	9.57	<.001
Mail × Home health	1.44	[1.07, 1.94]	2.40	.017
Mail × Outpatient clinic	1.03	[0.80, 1.33]	0.20	.844
Mail × Pediatric hospital	1.09	[0.79, 1.51]	0.53	.593
Mail × Rehabilitation hospital	1.06	[0.78, 1.44]	0.38	.706
Mail × Skilled nursing	0.83	[0.63, 1.10]	-1.32	.188

Using Web and General Hospital as the reference categories, the main effect of Mail had an odds ratio of 2.50 (p < .001), indicating substantially higher odds of response under mail in the reference facility type. Among the interaction terms, only Mail × Home Health was statistically significant (OR = 1.44, 95% CI [1.07, 1.94], p = .017), suggesting that the mail advantage was especially strong in home health settings relative to general hospitals. The other interaction terms

were not statistically significant, implying that most facility-specific deviations from the general mode effect were limited.
A practical implication is that ASHA may not need to use the same mode strategy across all facility types. Mail should be prioritized in settings where its advantage is strongest, especially home health, while lower-cost web-first or mixed-mode approaches may be sufficient in facility types where the mail advantage is smaller.

Subgroup 4: states and census regions

For the geographic analysis, states were recoded into 4 Census regions: Northeast, Midwest, South, and West.

Response rates were again higher under Mail than Web in all four Census regions, with a highly consistent mail advantage ranging from 14.4 to 15.6 percentage points. Chi-square tests were significant in all four regions (all $p < .001$), suggesting that the mode effect was robust across geographic contexts.

Table 14. Response Rates by Census Region and Mode

Region	RR Mail	RR Web	Δ (Mail − Web)
Midwest	30.5%	15.5%	+15.0%
Northeast	28.9%	13.3%	+15.6%
South	25.9%	11.5%	+14.4%
West	28.2%	13.2%	+15.0%

However, the region interaction test was not significant (ΔDeviance = 1.16, df = 3, $p = .762$), indicating that the effect of mode on response was consistent across the four Census regions. From a practical perspective, these findings suggest that region-specific mode strategies are likely unnecessary, since the same general mail-versus-web pattern appears throughout the country.

Recommendations from 2025 Survey

The most practical recommendation is to keep mail in the survey design, since it consistently outperformed web across almost all subgroups. A fully web-based approach would risk underrepresenting some groups, especially part-time employees and respondents in home health facilities, where the mail advantage was strongest. By contrast, separate mode strategies by job function or region do not appear necessary, because the mail effect was broadly similar across those groups. A targeted mixed-mode strategy would therefore be the most practical approach: use mail where it provides the greatest benefit for representation and response, and consider web-first approaches only where the mail advantage is smaller and cost savings are important.

3. Overall Recommendation

The optimal design is a **mail-first, limited follow-up strategy**:

1. Prioritize postal mail as the primary mode
2. Use web as a secondary option rather than a replacement
3. Implement a single, well-designed follow-up postcard
4. Focus on reducing respondent burden rather than increasing contact frequency

This approach balances response rate improvement with cost efficiency while minimizing differential nonresponse across key subgroups.

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Appendix

Codebook- https://github.com/Sagnik-Chakravarty/ASHA